Southern Cross University

Pharmacology of the Renal system

Diuretics

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Objectives for the renal system

- **Describe the structure and function of the portions of the renal system relevant to diuretics.**
- **Identify and explain the actions interactions and side effects of diuretics.**

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Medications of the Cardiovascular/Renal System:

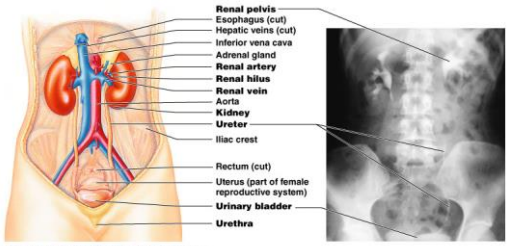
Diuretics

Lesson 1:

Understanding the foundational information about diuretics

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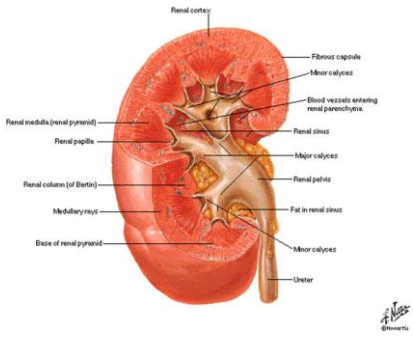
Location of the kidneys, ureters and bladder



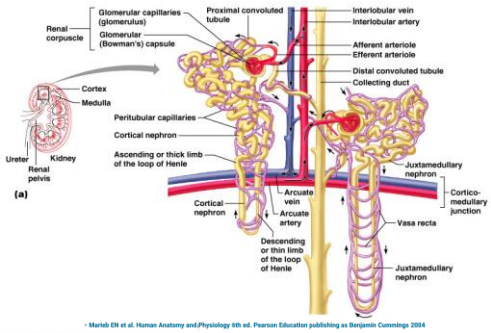
- Marieb EN et al. Human Anatomy and Physiology 6th ed. Pearson Education publishing as Benjamin Cummings 2004

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Renal pyramids

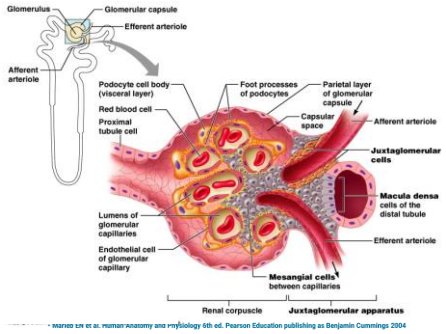


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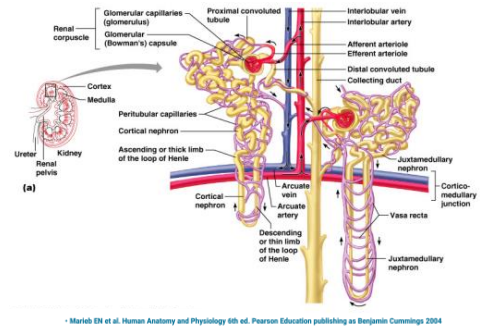
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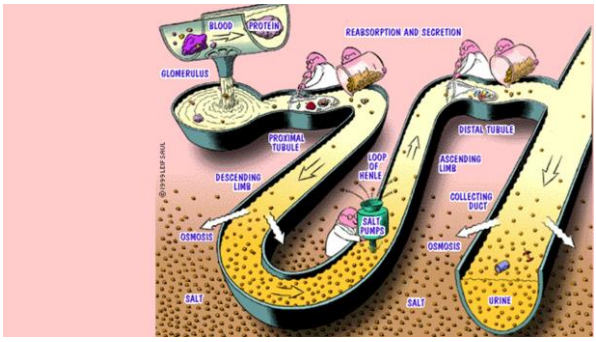
Medications of the Cardiovascular/Renal System:

Diuretics

Lesson 2:

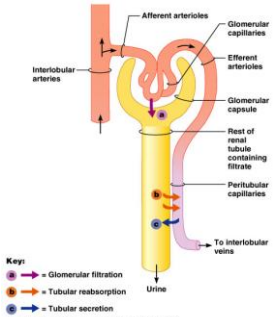
Understanding the foundational information about diuretics (continued)

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Filtration, reabsorption and secretion



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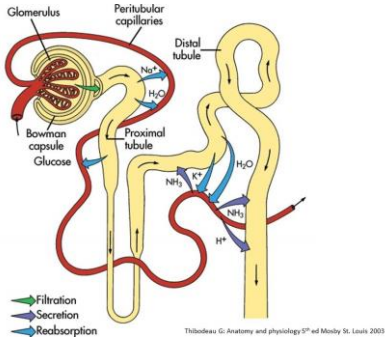
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Reabsorption and Secretion

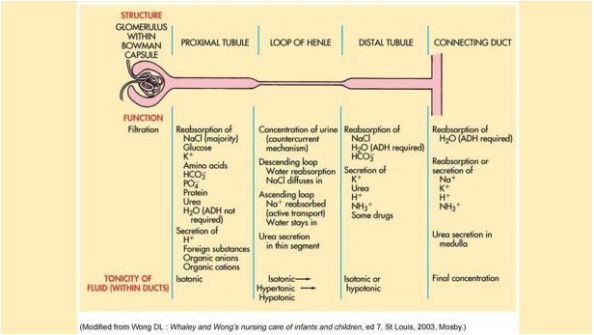
Reabsorption and secretion are confusing because we are talking about the tubule but the terms apply to the what is happening in relation to the peritubular blood supply.

- **Reabsorption of water and solutes by tubular cells into the peritubular capillary blood**
- **Secretion from the blood into the tubular lumen**

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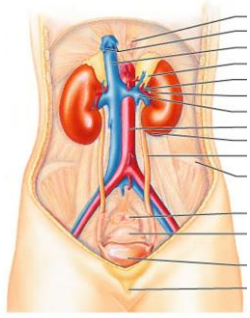


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Flow of urine



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Patients commencing diuretics

- **Care must be taken to avoid fluid rebound.**
- **Some patients decrease water consumption when they commence diuretics to decrease trips to the bathroom**
- **If water is decreased at the same time as diuretic is commenced, rebound oedema will result!**

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Medications of the Cardiovascular/Renal System:
Diuretics
Lesson 3:
Thiazide and thiazide-like diuretics

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Thiazide diuretics (hydrochlorothiazide)
Thiazide like diuretic (metolazone)

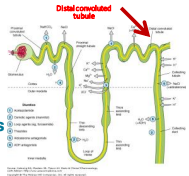
- Inhibit sodium reabsorption from the distal tubule
- Wastes potassium as well
- Used for hypertension, congestive heart failure, nephrolithiasis (kidney stones) and diabetes insipidus
- Thiazide diuretics compete with the secretion of uric acid so they are contraindicated in people with gout

- Several drugs use hydrochlorothiazide in combination – a little bit increases the action of other
  - Hydrochlorothiazide
  - 'thiazide" drugs

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Thiazide toxicity

- Allergic reaction (sulphonamides used in antimicrobial agents)
- Photosensitivity (skin; light pinkish skin are most sensitive)
- Hypo**k**alaemia (low potassium)
- Hypo**na**traemia (low salt Na+)
- Hyper**uric**aemia (gout)
- Hydrochlorothiazide
  - (Dithiazide)
- May increase risk of type 2 diabetes
- May increase lipid levels
- Contraindicated in pregnancy



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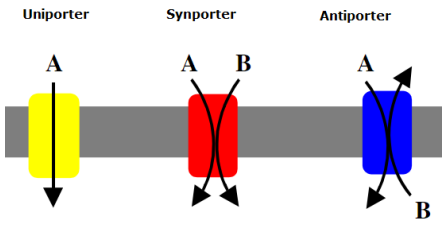
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Thiazide diuretics (hydrochlorothiazide)

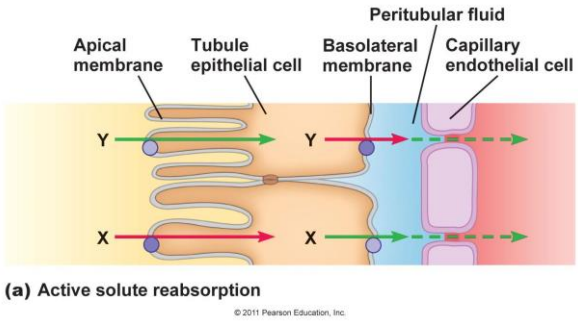
- May be indicated as an additional therapy for people with osteoporosis
  - Increases osteoblastic cells
  - Increases calcium levels.

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A symporter on the cell surface



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(a) Active solute reabsorption

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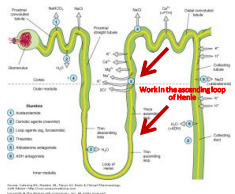
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Medications of the Cardiovascular/Renal System:
Diuretics
Lesson 4:
Loop diuretics

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Loop diuretics

- Furosemide (Lasix) and other loop diuretics work on the ascending loop of Henle
- They indirectly block active transportation of sodium, potassium and chloride into the interstitial space where they would normally re-enter circulation
- Sodium rich urine

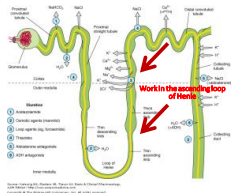


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Loop diuretics

- Produce a loss of fluid of up to 10 L/d
- Drugs of choice when rapid diuresis is needed
- Oral, IV or IM
- Alkalosis is possible
- Calcium is lost



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Loop diuretics

- Furosemide is Lasix (lasts six hours)
- Lasix prevents the reabsorption of:
  - Sodium
  - Potassium
  - Calcium
  - Magnesium
- Therefore, monitor for
  - Hypovolaemia and hyponaturemia
  - Hypokalemia
  - Hypocalcaemia
  - Hypomagnesemia
  - Also, allergies to those allergic to sulphonamides

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Medications of the Cardiovascular/Renal System:

Diuretics

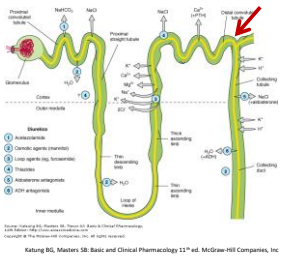
Lesson 5:

Potassium sparing diuretics

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Potassium sparing diuretics

- Blocks aldosterone at the distal tubule
- Spironolactone *(know this drug)*
- *OR*
- *Indirectly blocks the NaCl pumps at the same spot*
- *Amiloride*



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Potassium-sparing diuretics

Spironolactone

- Blocks aldosterone at the distal tubule
- Used for any aldosterone excess states.
- Toxicity includes hyperkalaemia (raised k+ potassium levels)
- Blocks androgen as well (dysmenorrhoea in women, breast development in males)

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Potassium-sparing diuretics

- Potassium-sparing diuretics
  - Spironolactone (Aldactone): Aldosterone antagonist
  - Amiloride: Blocks sodium channels
- Spironolactone was at one time thought to be important for all patients with hypertension but it causes hyperkalaemia
- Some say it should not be used with ACE inhibitors, but some people can tolerate it

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Medications of the Cardiovascular/Renal System:

Diuretics

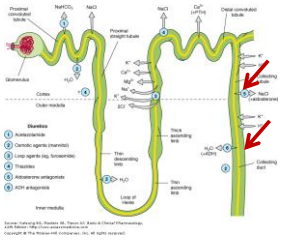
Lesson 6:

Other diuretics

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ADH antagonists

- "...vaptan" drugs
  - Blocks ADH receptors
- ETOH (alcohol)
  - Inhibits pituitary secretion of ADH
  - Not used as a diuretic but this explains why one is dehydrated after intoxication



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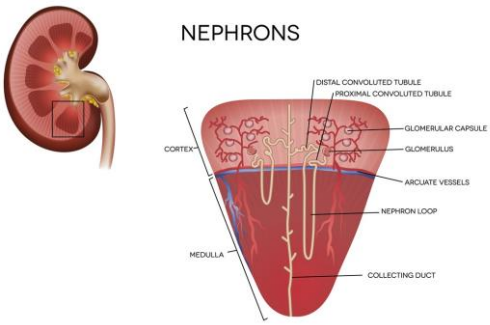
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Osmotic diuretics

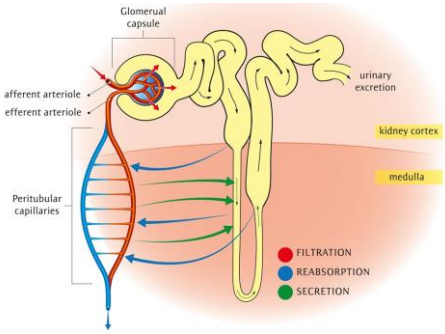
- Non-reabsorbable solute pulls water in
- Used to maintain urine volume and reduce intracranial and intraocular pressure

- Initially expands fluid volume (bad for CHF)
- Then high sodium because sodium is not following
- Produces hyperkalaemia

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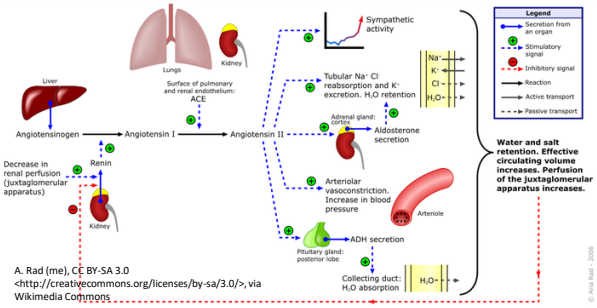
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ADH and aldosterone

- **ACE inhibitors act as diuretics by indirectly decreasing the ADH and aldosterone secretion**
- **Despite this, ACE inhibitors are not considered diuretics (they are in a class of their own and have diuretic actions)**
- **Some of the other diuretics act on**
  - **ADH the water saver and**
  - **aldosterone the salt saver**

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Renin-angiotensin-aldosterone system



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References

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